ELEC40006 Electronics Design Project

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1 Sensors

1.1 Radio Detection

The rock transmits its age using a radio signal with an 89 kHz carrier frequency. The signal uses two-level amplitude-shift keying, meaning the carrier is switched on and off to represent digital 1s and 0s. The data is sent at 600 bits per second as ASCII characters inside UART packets, with each character framed by 1 start bit and 1 stop bit. Each age message is four characters long and begins with #; for example, #123 represents an age of 1.23 billion years.

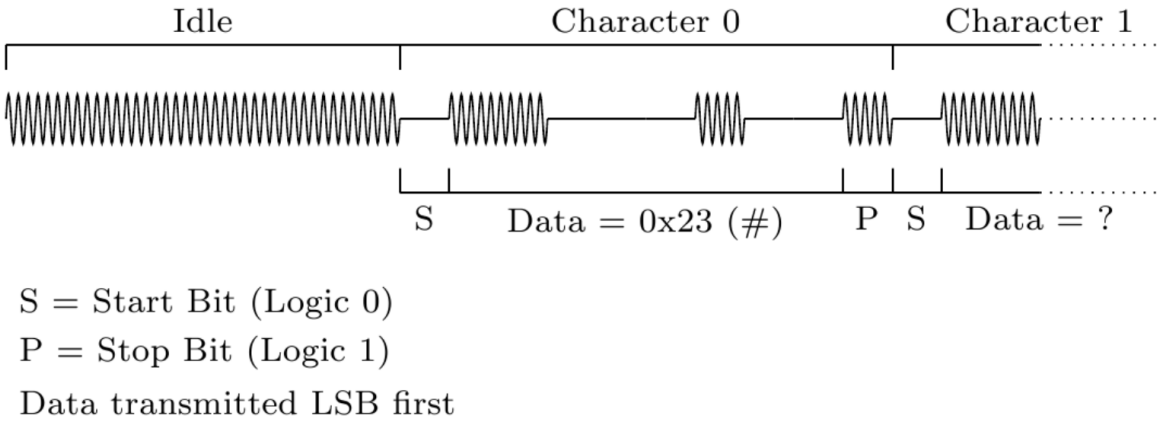


Figure 1: Radio Signal Waveform

To determine the specific age of each rock, the rover must intercept and decode a telemetry signal transmitted at a carrier frequency of 89 kHz, modulated using two-level amplitude-shift keying (ASK). This age metric is a core deliverable, requiring the software team to parse the decoded ASCII characters into a UART packet. To ensure robust signal detection, the receiving subsystem must address the following technical challenges:

- Filtering: Attenuate environmental noise by resonating strictly at the 89 kHz target frequency.
- Signal Amplification: Amplify low-amplitude induced signals to a usable voltage range while maintaining a high signal-to-noise ratio (SNR).
- Signal Conditioning: Rectify, smooth, and digitise the analogue waveform so the microcontroller can reliably process the 600 bits per second (bps) UART stream.



Figure 2: High-level block diagram illustrating the sequential signal processing stages of the radio detection architecture, from the initial coil antenna interception to the final digital UART input.

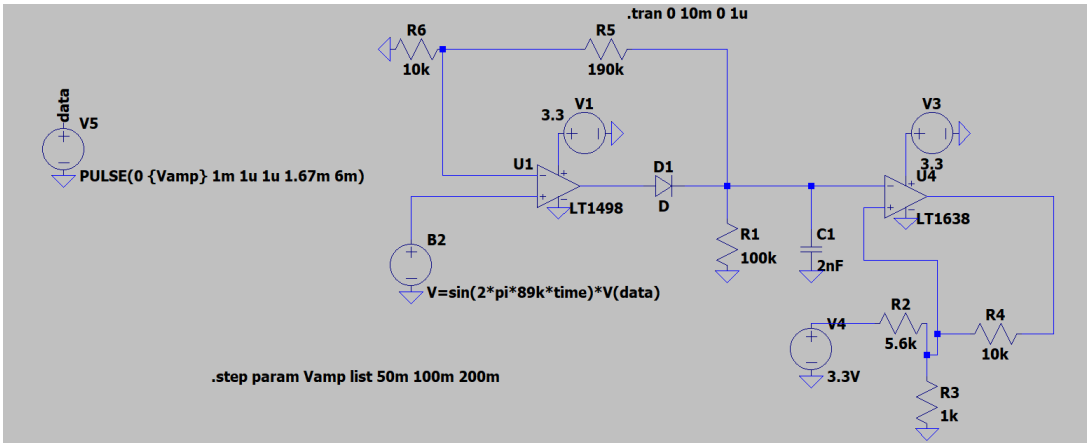


Figure 3: Complete schematic of the radio detection subsystem, detailing the active bandpass filtering, envelope detection, and Schmitt trigger digitisation stages.

1.1.1 Antenna Design

The primary parameters considered during the antenna design phase were the loop diameter and the total number of turns. Analytically, the relationship between these physical dimensions and the induced electromotive force (EMF) is governed by Faraday's Law of Induction:

$$V = -N\frac{d\phi}{dt}$$

Therefore, a larger coil area or a greater number of turns yields a higher induced voltage, significantly simplifying signal detection. However, increasing either variable introduces a direct trade-off by adding physical mass. To ensure the rover remains strictly under the 750g weight limit required to safely traverse the lunar surface weight-sensitive zone, a smaller, lightweight coil diameter was ultimately selected. To compensate for the reduced signal amplitude, active amplification stages were integrated into the circuit design, and the rover's navigation algorithm will be tasked with manoeuvring physically closer to the signal source.

1.1.2 Tuning Capacitor

To isolate the target 89 kHz carrier wave, an air-core inductor was custom-wound and its inductance was empirically measured. To achieve maximum voltage sensitivity at this specific frequency, the inductor was paired with a tuning capacitor to create a parallel LC resonant tank circuit. The required capacitance was calculated by rearranging the standard resonant frequency formula:

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \Rightarrow C = \frac{1}{4\pi^2 f_0^2 L}$$

Given the measured coil inductance (L) of 96 µH and the target resonant frequency (f0) of 89 kHz, the ideal capacitance was calculated to be approximately 33.3 nF. To match this requirement as closely as possible using standard off-the-shelf component values, a 33 nF capacitor was selected for the physical implementation.

1.1.3 Rectification

Following frequency isolation, the AC signal must be rectified. Several rectification topologies were evaluated against the constraints of varying operational distances and the low-amplitude nature of the induced signal. The findings are summarised in Table 2.1.1 below.

Table 2.1.1: Rectification Topology Decision Matrix

Topology Minimum Gain Stage Conclusion Signal Integration Required

Standard Diode ~0.7 V Poor Rejected. The 0.7 V threshold is (Half-Wave) too high to detect weak signals at a distance.

Schottky Diode ~0.2 V Poor Rejected. While an improvement, Rectification the 0.2 V drop may still clip the weakest telemetry signals.

Precision 0V Excellent Selected. Eliminates the diode Active Rectifier voltage drop entirely and easily accommodates an operational amplifier gain stage.

1.1.4 Envelope Detection

Once rectified, the 89 kHz carrier wave must be stripped away to reveal the underlying 1-bit ASK envelope. A passive RC envelope detector was implemented, with the resistor and capacitor values selected specifically to balance the carrier period and the bit transmission rate.

- The frequency of the carrier wave is 89 kHz, resulting in a period (T_c) of 11.2 µs.
- The data transmission rate is 600 bps, resulting in a bit period (T_{bit}) of 1.67 ms.

To satisfy the operational condition $T_c \ll \tau \ll T_{bit}$, component values of R = 100 kΩ and C = 3 nF were chosen, yielding an RC time constant (τ) of 300 µs. Because 300 µs is significantly larger than 11.2 µs, the capacitor discharges slowly enough to hold the signal HIGH between high-frequency peaks, effectively removing the ripple. Conversely, because 300 µs is much smaller than the 1.67 ms bit period, the capacitor has ample time to drain fully to 0 V when the rock stops transmitting a "1". This prevents the voltage from bleeding into the next "0" bit.

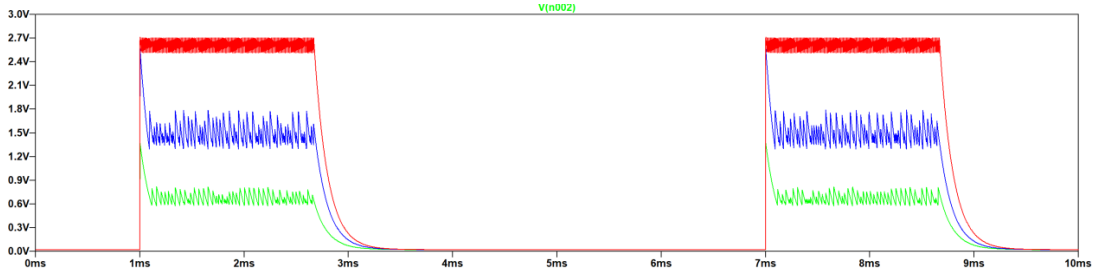


Figure 4: LTspice transient simulation at Point A. The multiple traces demonstrate the circuit's envelope detection response to an 89 kHz carrier wave stepped at varying input magnitudes (50 mV, 100 mV, and 200 mV), successfully smoothing the high-frequency signals into distinct DC pulses representing a logical '1'.

1.1.5 Schmitt Trigger

While the envelope detector successfully smooths the signal, the resulting waveform contains finite transition times and residual analogue ripple. To ensure reliable UART decoding by the software team, this signal must be strictly digitised. A Schmitt trigger was implemented to convert the analogue envelope into a crisp, logic-level square wave.

This design choice was highly cost and space-efficient, as it utilised the remaining free operational amplifier within the dual op-amp package.

The circuit uses an inverting topology, operating on a 3.3V supply. By utilising a positive feedback resistor network (comprising 27.5 kΩ, 349 Ω, and a 10 kΩ feedback resistor), the circuit establishes two distinct switching thresholds to provide noise immunity:

- Lower Threshold (VTL): Approximately 40 mV.
- Upper Threshold (VTL): Approximately 150 mV.

Finally, a 100 nF decoupling capacitor was placed in parallel with the reference voltage divider. This ensures that high-frequency noise from the power supply does not destabilise the precisely calculated switching thresholds.

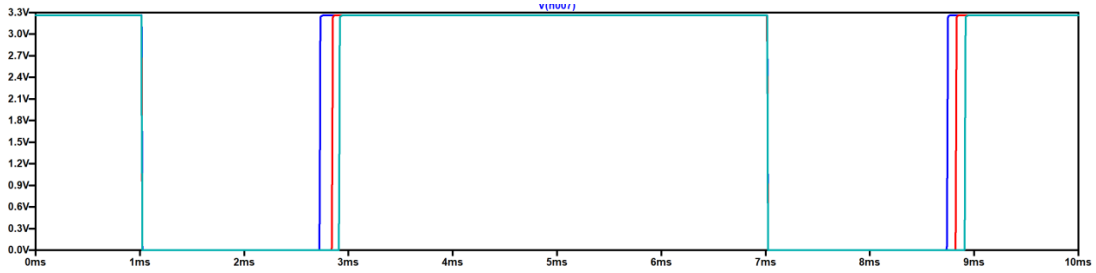


Figure 5: LTspice transient simulation at Point B. The output demonstrates the Schmitt trigger's response to the envelope signals derived from varying input magnitudes (50 mV, 100 mV, and 200 mV). The traces show a clean, robust 3.3V digital square wave, completely free of analogue ripple, ensuring reliable logic-level transitions for the MCU's UART serial pins.

1.2 Ultrasonic Rock-Type Detection Subsystem

The ultrasonic subsystem converts the presence of a 40 kHz signal emitted by a rock simulator into a single 3.3 V digital classification bit for the rover controller. The project guide states that rock type can be indicated by ultrasonic signal presence or absence at 40 kHz; it also specifies that the Metro M0 logic pins must not exceed 3.3 V, with logic-low at 0 V and logic-high at 3.3 V. The design therefore prioritised three requirements: reliable sensitivity to an attenuated 40 kHz signal, rejection of noise and non-ultrasonic content, and a microcontroller-safe digital output.

1.2.1 Requirements and Design Process

The design was developed as a signal-conditioning chain rather than as a software-sampled ultrasonic receiver. This choice was made because the received carrier amplitude is small after passing through the rock shell, while the required output is only binary. The signal chain is:

1. **Transducer pickup:** an MCUSD16A40S12RO ultrasonic transducer receives the rock signal at its 40 kHz resonance.
2. **AC coupling:** a 100 nF capacitor blocks any DC offset produced by the transducer before amplification.
3. **Band-pass gain:** an active band-pass stage centred at 40 kHz amplifies the desired carrier while attenuating low-frequency and broadband noise.
4. **Additional gain:** a second inverting gain stage increases sensitivity to the millivolt-level received signal.
5. **Envelope detection:** a passive rectifier and RC network converts the 40 kHz carrier into a slowly varying voltage proportional to received amplitude.
6. **Comparator:** a low-threshold comparator converts the envelope into a clean digital output: HIGH for ultrasonic signal present and LOW for absent.

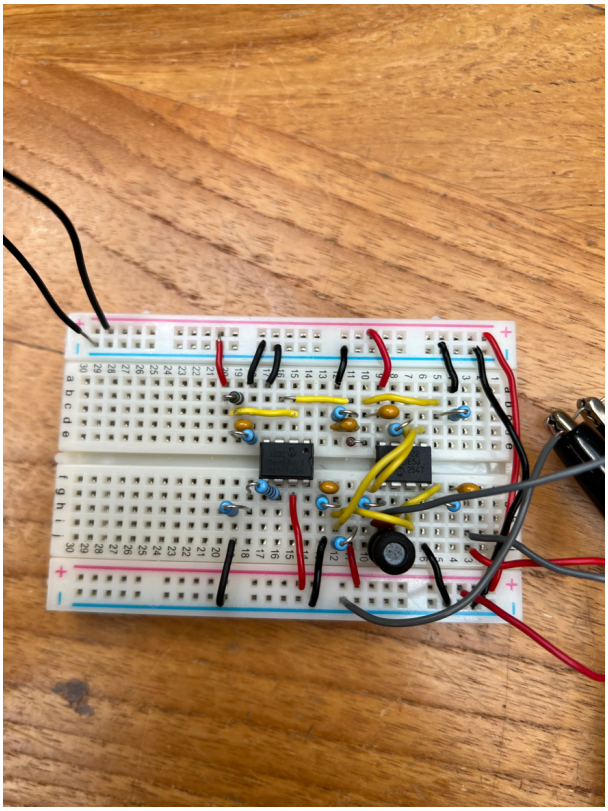


Figure 6: Breadboard Prototype

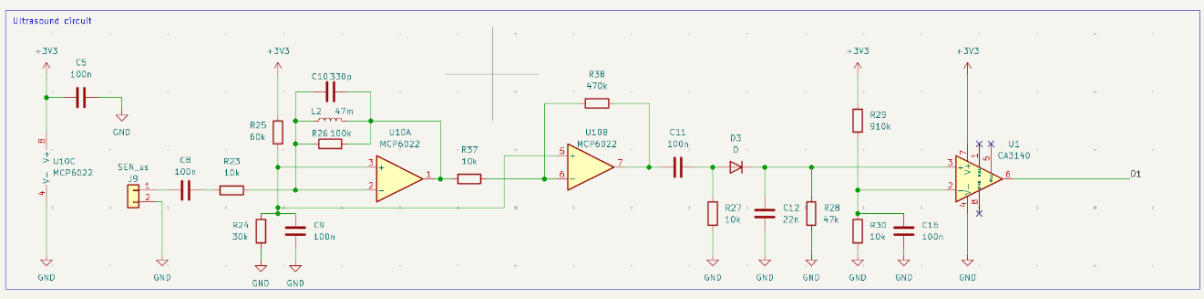


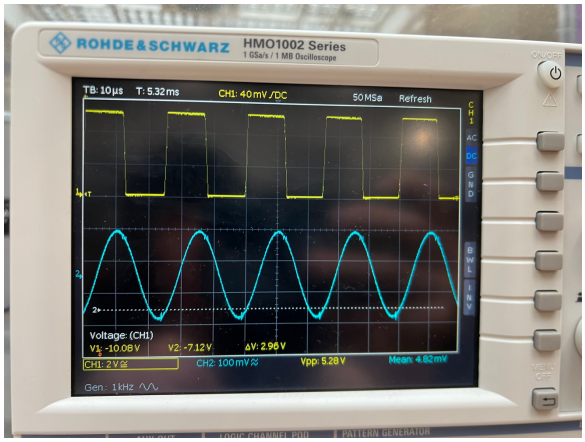
Figure 7: KiCad signal-chain schematic

The initial mechanical concept was to mount the ultrasonic transducer on a servo and actively aim it towards the rock drill hole. Testing showed that this would add mechanical complexity without being necessary: adding a further electronic gain stage allowed the transducer to detect the signal when placed near and facing the rock. This was a better design trade-off for an integrated rover because it reduced moving parts, control complexity, and alignment risk, while still providing the binary information required for rock classification.

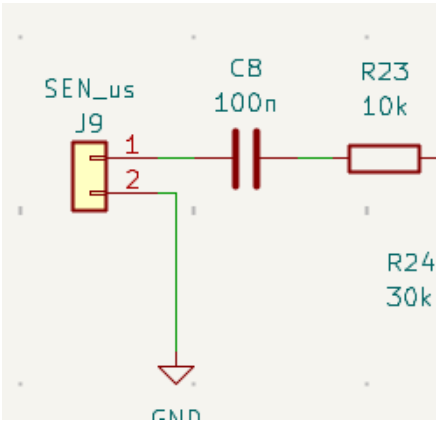
The supply strategy was also revised during prototyping. A 5 V supply gave stable readings up to roughly 0.5 m, but this range was excessive for the final arena because nearby rocks could produce false detections. The subsystem was therefore moved to 3.3 V operation. This made the output directly compatible with the Metro M0, reduced level-shifting requirements, and intentionally biased the design towards local proximity detection rather than long-range detection.

1.2.2 Implementation

Input coupling and filtering. Without the rock shell, the received signal was measured at about 300 $mV_{p\text{-}p}$. With the shell present, the signal was attenuated by approximately a factor of 100 to around 3 mV to 5 mV. The input therefore required high gain, but not broadband gain: amplifying all frequencies would increase susceptibility to electrical noise and other signals in the rover environment.



(a) Transducer signal measurement.



(b) 100 nF AC-coupling capacitor.

Figure 8: Front-end evidence: the transducer output is small enough after shell attenuation to require active gain and filtering.

An inverting active band-pass topology was selected using the MCP6022 dual op-amp. Its peak gain is set approximately by the feedback-to-input resistance ratio, but the feedback resistance also changes bandwidth and quality factor. The band-pass stage gain was therefore limited so that the filter retained a practical bandwidth of approximately 5 kHz and a quality factor near 8 around the 40 kHz carrier. The finite impedance of the practical LC network at resonance, measured at roughly 100 kΩ, also constrained the design: the input impedance needed to remain at least around 10 kΩ to avoid loading the transducer, so the first-stage gain was kept below about 10.

A separate inverting amplifier was added after the band-pass stage. Combined with the band-pass stage, the implemented gain was approximately 470. Because the subsystem only detects whether the carrier is present, waveform distortion at high signal levels was acceptable; the important metric was reliable separation between noise and a true ultrasonic signal. Both gain stages were biased at approximately 1.1 V, below the 3.3 V midpoint, giving more positive headroom for the envelope detector that follows.

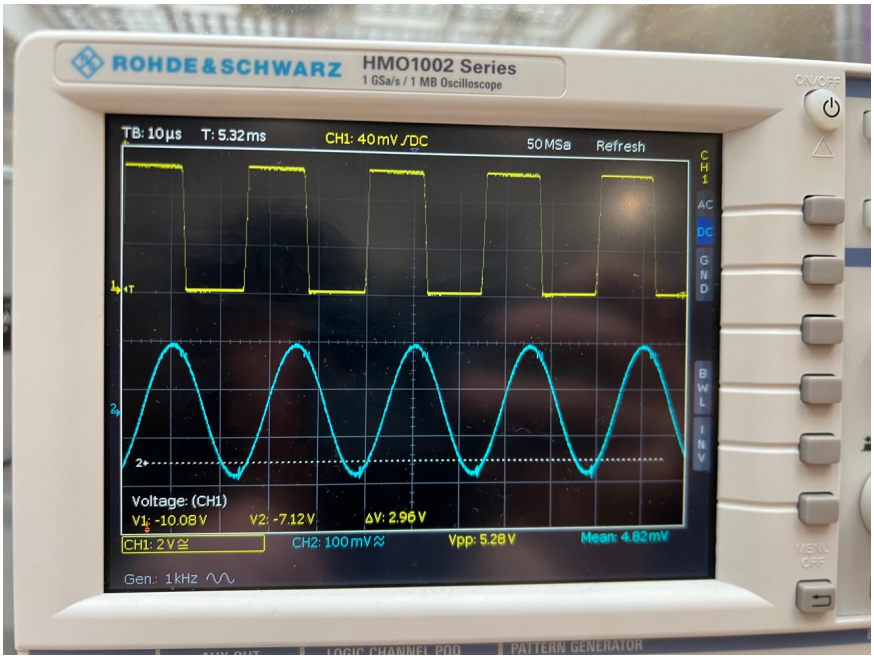
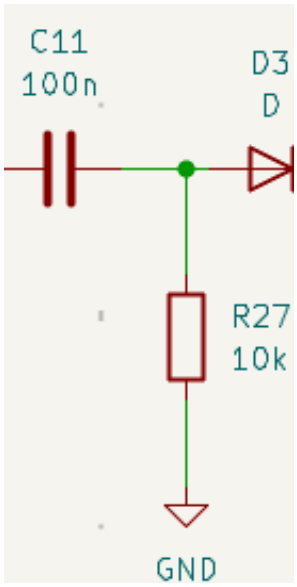


Figure 9: Measured gain-stage behaviour. The overall gain was documented as approximately 470, sufficient to lift the shell-attenuated millivolt input into a detectable range.

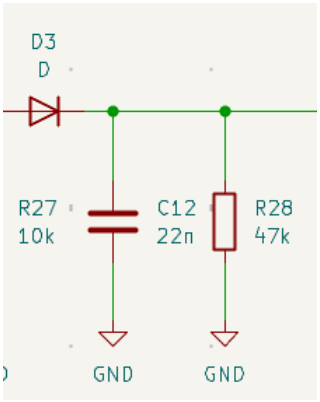
Envelope detector. After amplification, the circuit removes the op-amp bias using a high-pass filter with a cutoff near 159 Hz. The resulting AC signal is rectified and smoothed by a passive RC envelope detector. The documented design used a 22 nF capacitor and a 10 kΩ discharge path, giving

$$\tau = RC = 10\,\mathrm{k\Omega} \times 22\,\mathrm{nF} = 220\,\mu\mathrm{s}. \tag{1}$$

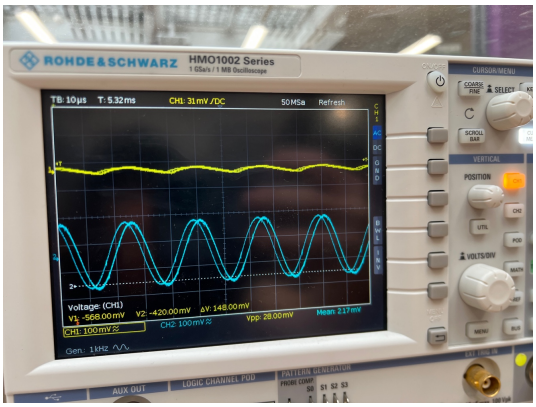
At 40 kHz, the carrier period is 25 µs, so this time constant is about 8.8 carrier periods. It is therefore long enough to smooth the carrier ripple while remaining short enough for the digital output to respond on a sub-millisecond scale. The schematic also includes a 47 kΩ load/discharge resistor, so the exact decay rate depends on the final assembled network; the key design condition is that the envelope time constant is much larger than the ultrasonic period.



(a) Bias-removal high-pass filter.



(b) Envelope detector.



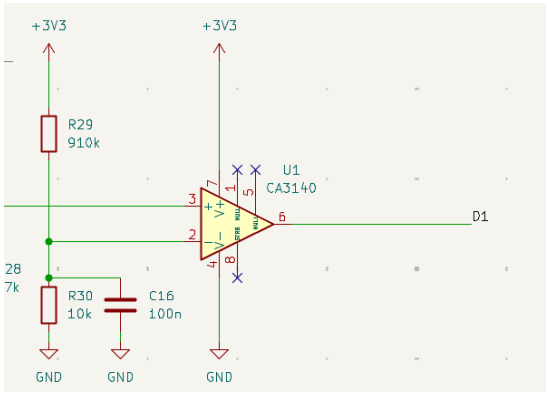
(c) Envelope output.

Figure 10: Envelope detector implementation. The oscilloscope evidence shows the carrier converted into a near-DC amplitude signal, with the expected diode voltage drop.

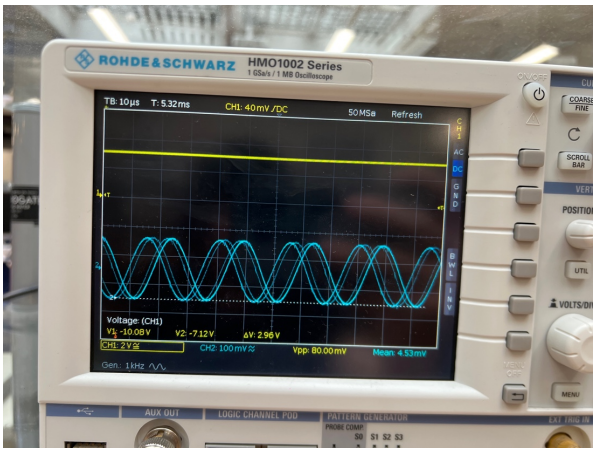
Comparator and digital interface. The envelope output was reported to range from about 100 mV to above 1 V when the ultrasonic signal was present. A threshold of approximately 36 mV was generated with a high-ratio divider from the 3.3 V rail:

$$V_{\text{th}} = 3.3\,\text{V}\frac{10\,\text{k}\Omega}{910\,\text{k}\Omega + 10\,\text{k}\Omega} \approx 35.9\,\text{mV}. \tag{2}$$

This threshold sits above the observed noise floor while remaining comfortably below the envelope amplitude produced by a true received signal. The comparator was implemented with an MCP6541 in the physical circuit; the KiCad schematic uses a CA3140 symbol because it was available in the library and shares the required pinout. The comparator output is interpreted directly by the microcontroller: HIGH at 3.3 V when the 40 kHz carrier is present, LOW at 0 V otherwise.



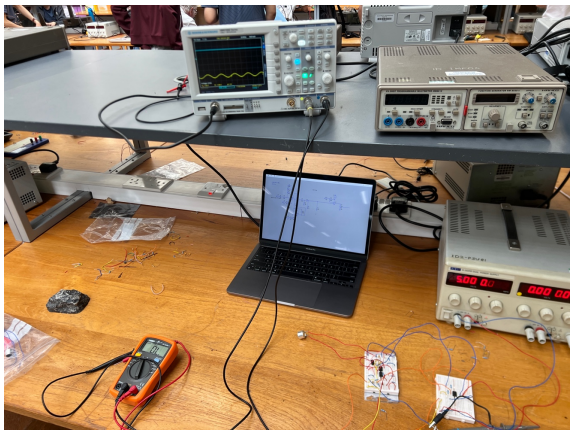
(a) Comparator threshold network and symbol.



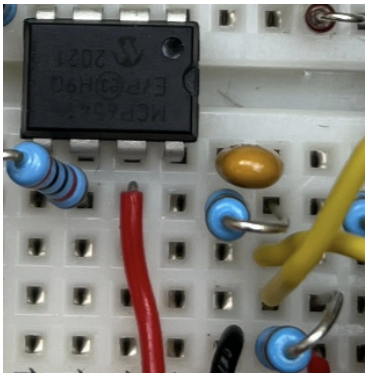
(b) Comparator output: HIGH when ultrasonic signal is present.

Figure 11: Comparator stage converting the analogue envelope to a Metro-compatible digital rock-type bit.

Decoupling capacitors were added to the power rails and bias nodes to reduce supply noise and bias drift. This is important because the circuit contains high gain before thresholding: any supply-coupled noise at the op-amp inputs can be amplified and create false comparator transitions.



(a) Bench setup used during range and supply-voltage trials.



(b) Prototype decoupling.

Figure 12: Integration-focused prototype refinements: supply selection and decoupling improved compatibility with the rover electronics.

1.2.3 Evaluation

The subsystem was evaluated at each stage rather than only at the final digital output. This is appropriate because failures in the transducer, gain/filter, envelope detector, or comparator can produce the same final symptom: an incorrect binary rock-type bit. The evidence collected is summarised in Table 1.

Table 1: Requirement-level evaluation of the ultrasonic detection subsystem.

Requirement	**Test or evidence**	**Measured/observed result**	**Outcome**
Detect the 40 kHz ultrasonic carrier	Oscilloscope measurement at the transducer and after amplification	The unshielded signal was about 300 $\mathrm{mV_{p\text{-}p}}$; with the rock shell it fell to about 3 mV to 5 mV, confirming the need for high gain.	Met
Reject unwanted frequencies and noise	Active band-pass design centred at the carrier frequency	Band-pass centred at 40 kHz, with approximate bandwidth 5 kHz and $Q \approx 8$. Gain of this stage was limited to preserve bandwidth and input impedance.	Met
Amplify the attenuated signal enough for thresholding	Cascaded active gain stages	Overall gain of approximately 470 produced envelope amplitudes from about 100 mV to above 1 V, well above the comparator threshold.	Met

Continued on next page

Requirement	Test or evidence	Measured/observed result	Outcome
Convert carrier amplitude to a slow analogue signal	Envelope detector oscilloscope trace	RC detector produced a near-DC envelope; the observed diode drop was approximately 0.7 V, consistent with a passive rectifier.	Met
Provide a clean microcontroller-safe digital bit	Comparator output test	Threshold was set to about 36 mV; output switched to 3.3 V when ultrasonic signal was present and 0 V when absent.	Met
Avoid false detections in dense rock placement	Supply-voltage and range trial	At 5 V, detection worked up to about 0.5 m; this was judged too sensitive for the demo environment, so the circuit was moved to 3.3 V to favour proximity detection and direct logic compatibility.	Improved; final arena test required
Maintain stable bias and supply rails	Prototype inspection and schematic update	Decoupling capacitors were added to rails and bias points to reduce false switching from supply or bias noise.	Met in prototype

The strongest result is the clear separation between the envelope amplitude when a 40 kHz signal is present and the 36 mV comparator threshold. Even the lower reported envelope value of approximately 100 mV gives a margin of about 64 mV, while stronger detections exceed 1 V. This margin explains why a single-threshold comparator was sufficient during bench tests and why the final output remained stable rather than flickering near the switching point.

The main limitation is that bench range is not the same as integrated rover performance. The 5 V version demonstrated sensitivity, but the later 3.3 V version was selected because excessive range could classify the wrong rock in a dense field. The final report should therefore include an integrated test in which the rover approaches multiple rocks at realistic spacing and records true positives, false positives, and missed detections against distance and angle. The most useful final metrics would be: maximum reliable detection distance, minimum spacing before cross-detection occurs, angular tolerance of the transducer, and digital-output stability over repeated trials.

Overall, the subsystem satisfies its core electrical requirements: it detects a weak 40 kHz ultrasonic signal, filters and amplifies it using a technically justified analogue signal chain, converts the carrier into an envelope, and outputs a safe 0 V/3.3 V binary signal for rock-type classification. The design process also shows meaningful iteration: the servo aiming mechanism was removed in favour of electronic gain, and the supply voltage was changed from 5 V to 3.3 V after evaluating the practical risk of false detections in the final demonstration environment.

1.3 Infrared Sensor

1.3.1 Role in the sensing system

The infrared subsystem estimates the random optical-event rate emitted by the rock simulator. This rate is one of the three properties used by the classification algorithm. Basaltoid and Lunarite produce the high infrared rate, $547\,\mathrm{s}^{-1}$, whereas Gravion and Regolix produce the low infrared rate, $312\,\mathrm{s}^{-1}$. Infrared sensing therefore distinguishes the two radioactivity classes, but

it cannot identify the rock type by itself; the final classification is obtained by combining the infrared result with the ultrasonic and magnetic decisions.

The design requirements follow from the rock specification and from the practical constraints of the rover electronics. The rock emits infrared events at one of two nominal rates, $312\,\mathrm{s}^{-1}$ or $547\,\mathrm{s}^{-1}$, with event arrivals modelled as a Poisson process. The technical guide specifies an optical wavelength of approximately 950 nm, pulse widths of about 50 µs, and possible interference from ambient light, particularly DC illumination and 100 Hz ripple from mains-powered lighting. The analogue front end must therefore detect weak, short optical pulses, reject slowly varying ambient-light components, and produce a clean microcontroller-safe output over the 0 V to 3.3 V range.

1.3.2 Poisson counting basis

The required measured quantity is the pulse rate, not the pulse amplitude. If N pulses are counted over a fixed observation window T, then

$$N(T) \sim \mathrm{Poisson}(\lambda T),$$

where λ is the event rate. Hence,

$$\mathrm{E}[N] = \lambda T, \qquad \sigma_N = \sqrt{\lambda T}.$$

For a 120 ms counting window, the expected counts are

$$\mathrm{E}[N_{547}] = 547 \times 0.120 = 65.6,$$
$$\mathrm{E}[N_{312}] = 312 \times 0.120 = 37.4.$$

A decision threshold placed halfway between these means gives

$$N_{\mathrm{th}} \approx \frac{65.6 + 37.4}{2} = 51.5.$$

The implemented rule is therefore to classify the signal as high-rate when $N \geq 52$, and as low-rate otherwise. Under ideal counting and nominal Poisson rates, this gives a false-high probability of approximately 1.4 % for the low-rate class and a false-low probability of approximately 3.6 % for the high-rate class.

Increasing the observation window improves reliability: the separation between the two expected counts increases linearly with T, while the Poisson standard deviation increases only as $\sqrt{T}$.

Table 2: Expected infrared-count separation for different software counting windows. The probabilities assume ideal pulse detection and only Poisson count variation, so they are a lower-bound estimate of the final system error.

Window T	Low-rate mean	High-rate mean	Count threshold	Approx. worst error
0.12 s	37.4	65.6	52	3.6 %
0.25 s	78.0	136.8	108	0.49 %
0.50 s	156.0	273.5	215	0.011 %

This statistical analysis justifies using an interrupt-driven pulse counter rather than estimating the instantaneous time between adjacent pulses. The mean inter-arrival time is only about 1.83 ms for the high-rate class and 3.21 ms for the low-rate class, while each optical pulse is only 50 µs wide. A fixed counting window is therefore more robust than a single-period measurement

because it averages over the random nature of the source. The counting rule can also be written as a binary hypothesis test on the Poisson rate,

$$H_0 : \lambda = 312\,\mathrm{s}^{-1}, \qquad H_1 : \lambda = 547\,\mathrm{s}^{-1}.$$

For a counting window of duration T, the observed count satisfies

$$N(T) \sim \mathrm{Poisson}(\lambda T).$$

A natural midpoint threshold in rate is

$$\lambda_{\mathrm{th}} = \frac{312 + 547}{2} = 429.5\,\mathrm{s}^{-1},$$

so the corresponding integer count threshold is

$$N_{\mathrm{th}}(T) = \lceil \lambda_{\mathrm{th}} T \rceil .$$

The decision rule is therefore

$$\widehat{H} = \begin{cases} H_1, & N(T) \geq N_{\mathrm{th}}(T), \\ H_0, & N(T) < N_{\mathrm{th}}(T). \end{cases}$$

For example, with $T = 1\,\mathrm{s}$, this gives

$$N_{\mathrm{th}}(1) = \lceil 429.5 \rceil = 430.$$

The false-high probability under the low-rate hypothesis is

$$P_{\mathrm{FH}} = \Pr[N \geq 430 \mid \lambda = 312] = \sum_{n=430}^{\infty} \frac{312^n e^{-312}}{n!} \approx 1.51 \times 10^{-10},$$

while the false-low, or missed-detection, probability under the high-rate hypothesis is

$$P_{\mathrm{FL}} = \Pr[N < 430 \mid \lambda = 547] = \sum_{n=0}^{429} \frac{547^n e^{-547}}{n!} \approx 9.21 \times 10^{-8}.$$

These probabilities are much smaller than those for the shorter software counting windows because the separation between the expected counts grows linearly with T, whereas the Poisson standard deviation grows only as $\sqrt{T}$. Hence, we chose T = 1 s as our window.

1.3.3 Initial Analogue Front-end Design

The signal chain used in the rover is:

$$\begin{aligned} &\text{Photodetector} \rightarrow \text{AC coupling / high-pass rejection} \rightarrow \text{MCP6022 gain stage} \\ &\rightarrow \text{comparator threshold} \rightarrow \text{GPIO interrupt input.} \end{aligned}$$

In a phototransistor, incoming photons provides energy to generate electron hole pairs in base, which converts into a base current. In the breadboard prototype this was implemented using a silicon phototransistor, with the emitter resistor converting photocurrent into a voltage. For a first-order approximation,

$$V_{\mathrm{sense}} \approx I_{\mathrm{photo}} R_E. \tag{3}$$

A phototransistor was preferred over a general visible-light detector because the rock signal is centred in the near-infrared. The initial sensor choice was SFH 300, taken off from the

EEEBug, which was suitable electrically but had a relatively narrow receiving angle, making the measurement sensitive to rover alignment.

The first iteration was built with a capacitor to filter out ambient noise (DC), non-inverting amplifier with a gain of 100, and U2 is a open collector comparator so it needs a pullup resistor to output 3.3V logic.

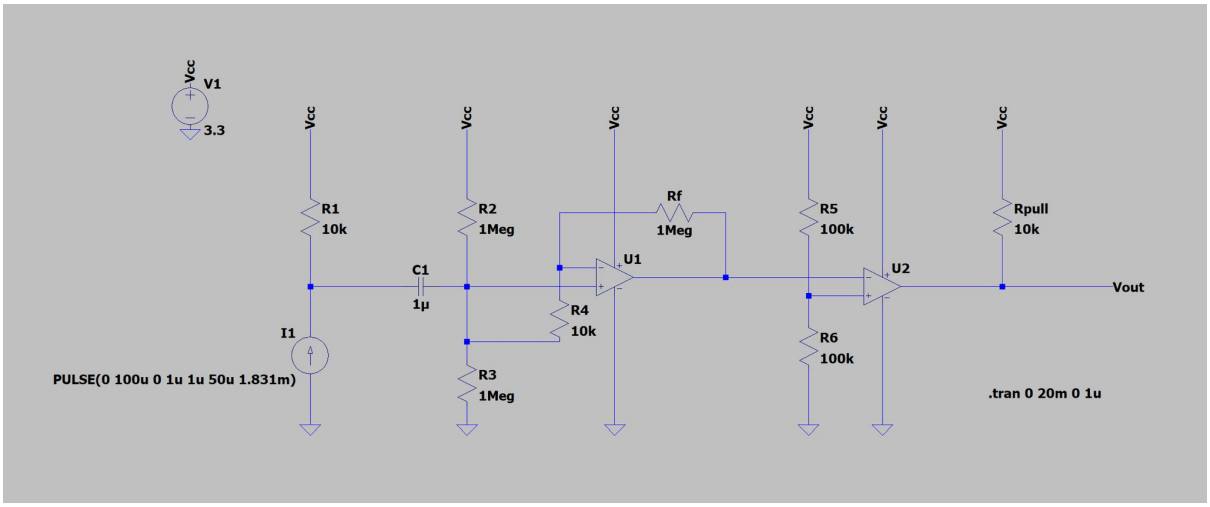
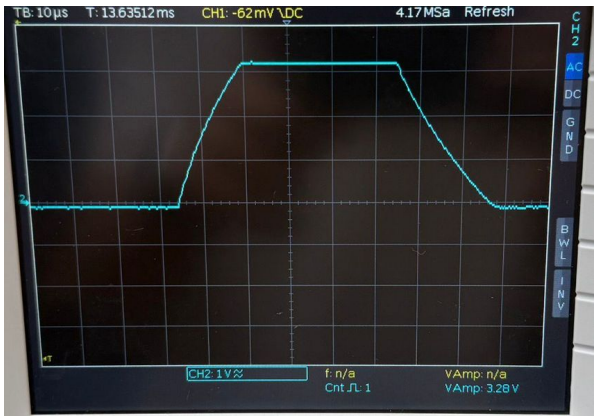
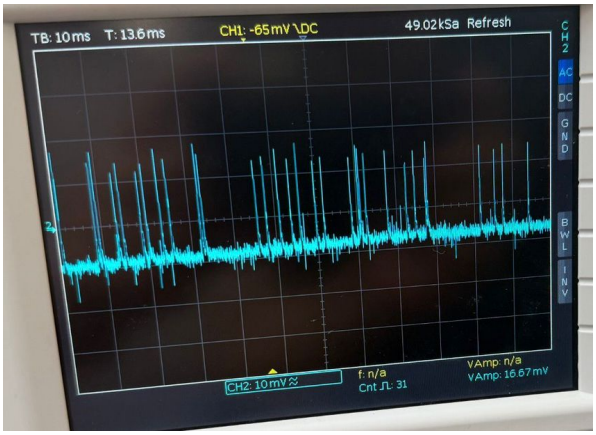


Figure 13: 1st Iteration finalized on 5/21

We quickly found out in testing that this was a non-functional design as it didn't filter out the ambient noise when the car was moving as it's variable, and was not able to produce a consistently signal.

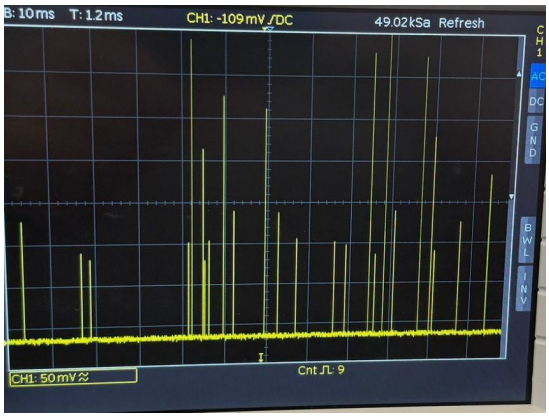


(a) Slewed Pulse

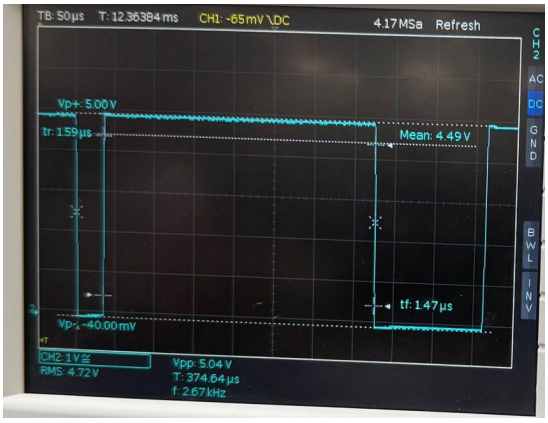


(b) Inconsistent Pulse Readings 1

Figure 14: Oscilloscope measurements.



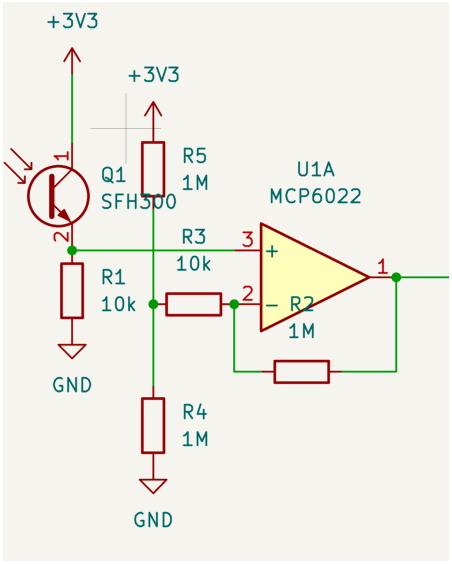
(c) Inconsistent Pulse Readings 2



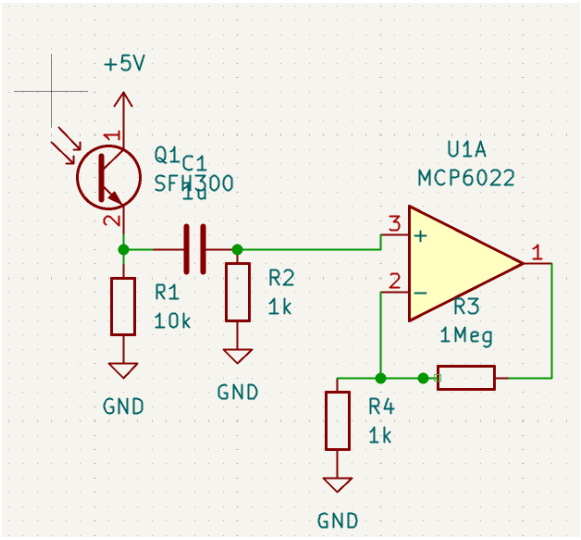
(d) Extended Pulse Lasting 350 µs

Figure 14: Oscilloscope measurements, continued.

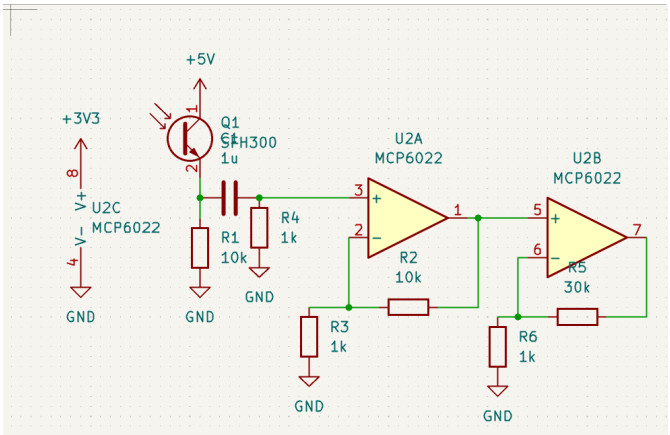
After a week of debugging with no progression, we decided to allocate more members to help out and redesign the circuit on 5/27.



(a) DC Bias, Collector to 3V3



(b) AC Coupling High Pass Filter



(c) Second Stage added

Figure 15: Phototransistor amplifier circuit variants.

We first tried using DC bias to remove any DC effect, but proved not useful due to the variable noise level which causes variable DC. We then came up with AC coupling using a high pass filter with time constant 1 ms, which is much larger than pulse width 50 µs. The corner frequency of the filter is:

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 1 \times 10^3} = 159\text{Hz},$$

which is much lower than the average frequency of the pulses.

A large time constant is chosen because the capacitor voltage barely changes during the pulse, maintaining its shape. This successfully eliminated the DC. We added a DC bias to retrieve the "lost volts" as the output would go into the negative as we hadn't set a DC bias. We were unable to recreate these conditions where the output would rest at a negative value.

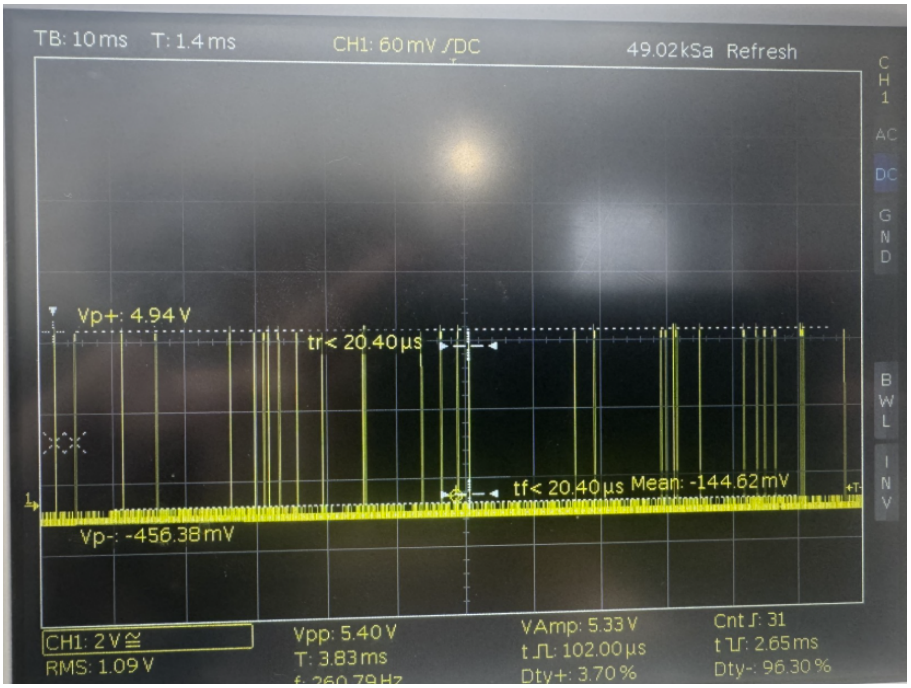


Figure 16: Noisy signal due to Johnson Effect

A high gain of 1000 also made Johnson effect non-negligible, which states that any component that passes current is a source of noise, where the mean square noise voltage is:

$$\overline{v_n^2} = 4k_BTBRV^2,$$

where T is temperature in Kelvin, B is bandwidth noise in Hertz, R is resistance in Ohms.

That means the resistance should be reduced to reduce the Johnson effect. Using small resistors is not a good choice because it dissipates a lot of current. Since we are using a dual channel opamp MCP6022, we came up with a 2 stage design with reduced resistance and gain in each stage. Since the gain $G = 60 \times 60 = 3600$, we also introduced an extra AC coupling between stages to remove the noise. We also applied decoupling capacitors for voltage stability, as capacitors can provide charge locally quickly.

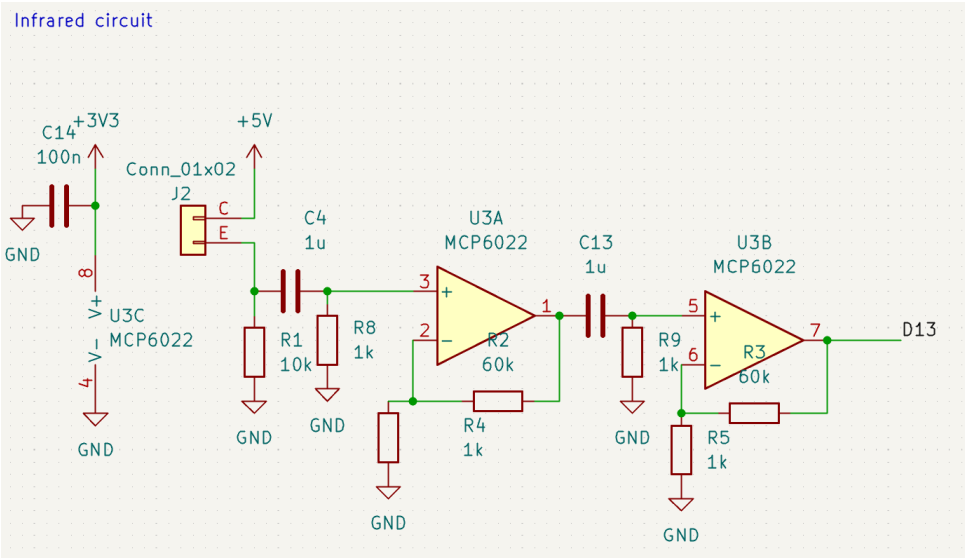


Figure 17: Final Iteration of Infrared Sensing Circuit

However, even after everything was optimized, the circuit was only able to detect the pulses when the PCB was removed, or the sensor is touching the rock at an particular angle. So we suspected that the sensor SFH 300 3/4 was not good enough and conducted a research, and decided to use SFA 314 FA.

Parameter	**SFH 300**	**SFH 314 FA**
Angle	50°	80°
Spectrum	Visible and IR	IR
Photocurrent at $V_{CE} = 5\,\mathrm{V}$, $S_{\mathrm{rel}} = 100\,\%$, $E_e = 1\,\mathrm{W\,cm^{-2}}$, bin 4	at least 3.2 mA	5.6 mA to 9.0 mA

Table 3: Comparison of SFH 300 and SFH 314 FA parameters.

It had a much wider angle, and a larger photocurrent, which means that it should be more sensitive and detect the pulses under less constraints. They come in single packages in different bins, with bin 1 producing the lowest photocurrent and 4 the higher. There are 2 purchse choices, either guaranteed bin 2/3, or completely random. Since the difference of SFH 314 FA 2/3 is not significant to SFH 300 3/4, we chose to buy 10 independent random packages for 3.43 £.

Photocurrent Groups

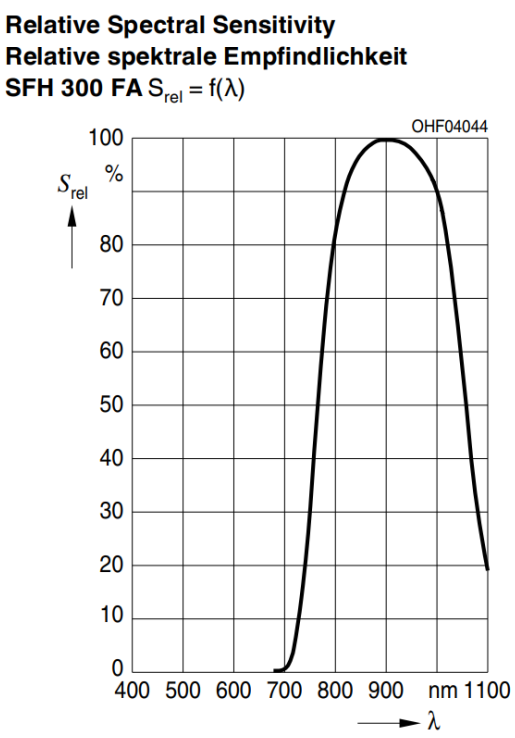
T_A = 25 °C

Group	Photocurrent [1] V_{CE} = 5 V; λ = 950 nm; E_e = 0.5 mW/cm² min. I_{PCE}	Photocurrent [1] V_{CE} = 5 V; λ = 950 nm; E_e = 0.5 mW/cm² max. I_{PCE}
1	0.71 mA	1.12 mA
2	1.12 mA	1.80 mA
3	1.80 mA	2.80 mA
4	2.80 mA	4.50 mA

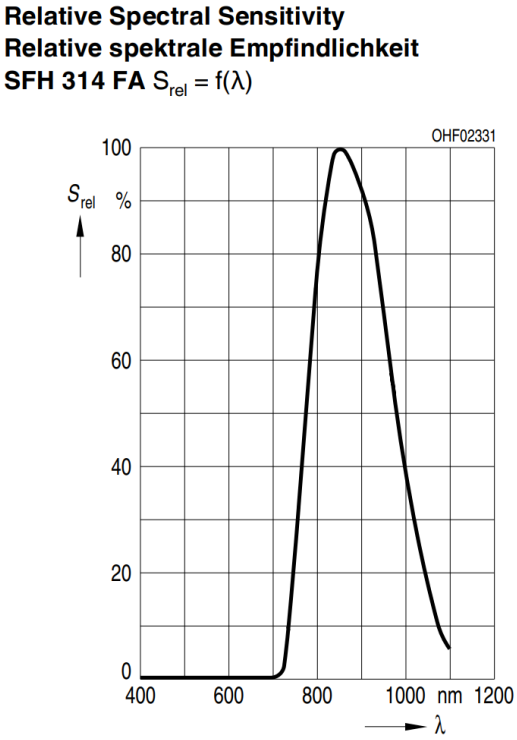
Figure 18: Photocurrent Groups for SFH 314 FA

Also a significant advantage was its sensitivity peaks at 950 nm, very close to 940 nm pulses.

The gradient is also very steep meaning that it would experience much less noise from the ambient.



(a) SFH 300 Relative Spectral Sensitivity



(b) SFH 314 FA Relative Spectral Sensitivity

Figure 19: Relative spectral sensitivity comparison

We were able to detect the pulse 3cm away from the rock using the new sensor.

1.3.4 Gain, thresholding and design iteration

The first circuit version used a photodetector, AC-coupling stage, amplifier and threshold stage, producing a 3.3 V digital output for the microcontroller. Testing showed two main issues. First, the photodetector output was weak when the rock cover attenuated the infrared intensity. Secondly, the threshold output polarity was inconvenient for software counting because the detected pulse appeared as a falling transition.

The second version addressed both issues. The feedback resistor was increased from 100 kΩ to 1 MΩ, increasing the nominal analogue gain from approximately ×10 to ×100. The MCP6022 was selected because it is a rail-to-rail input/output op-amp with a typical gain-bandwidth product of 10 MHz, which is sufficient for amplifying short infrared pulses at the chosen gain while operating from the rover's low-voltage supply [**?**]. The comparator inputs were then swapped so that each detected infrared event generated a positive-going digital pulse at the microcontroller input, simplifying interrupt-based counting.

The use of a comparator after amplification is important because the software does not need to know pulse amplitude. It only needs a reliable transition for each optical event. The comparator threshold therefore separates real pulses from residual noise and produces a signal suitable for a GPIO interrupt. A short software refractory time after each interrupt can prevent one physical pulse from being counted multiple times if the analogue waveform rings around the threshold.

The Metro M0 input protection constraint also shaped the circuit. The project guide warns that the Metro M0 logic pins must not be driven above 3.3 V [**?**]. Therefore the threshold stage

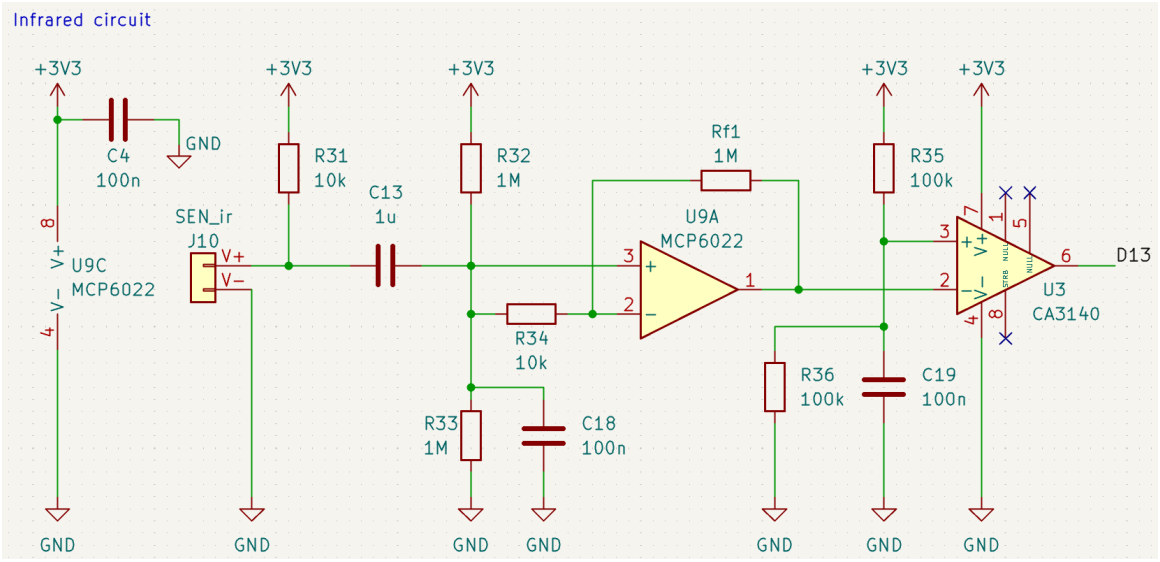


Figure 20: Second-version infrared receiver circuit. The MCP6022 stage amplifies the AC-coupled pulse component, while the comparator stage converts the analogue pulse into a logic-level signal for the microcontroller input.

is designed to output logic compatible with the 0 V to 3.3 V input range, rather than passing a higher-voltage analogue signal directly into the microcontroller.

1.3.5 Testing and evaluation

Bench testing confirmed that the analogue front-end could produce visible, countable pulse transitions on the oscilloscope. Figure 21 shows the amplified pulse train observed during development. The aim of this test was not to recover the exact optical waveform, but to verify that the pulse edges remained distinct enough for threshold counting.

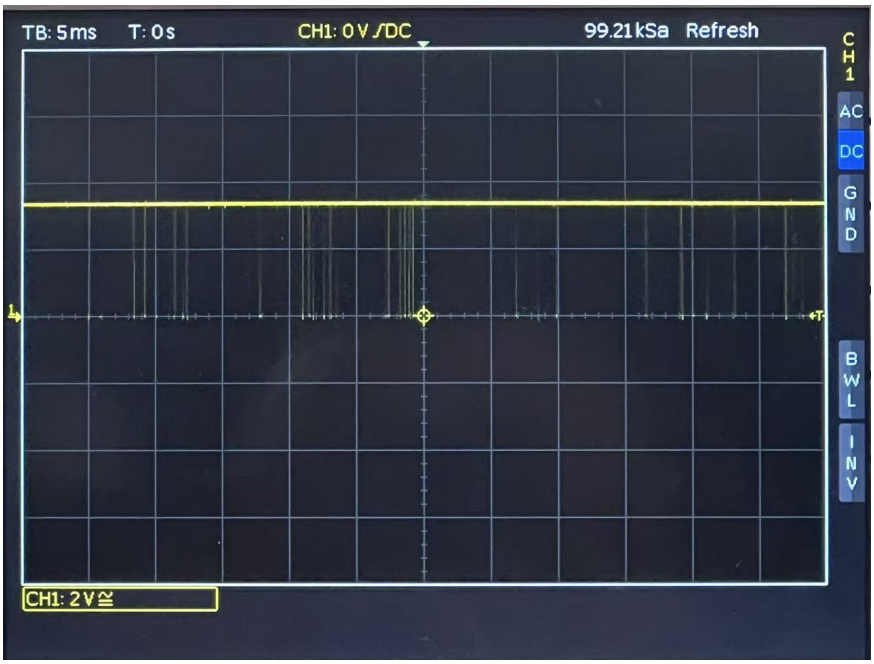


Figure 21: Oscilloscope capture of amplified infrared pulses during prototype testing. Countable pulse clusters are visible, supporting the decision to use a threshold-counting measurement.

The final subsystem-level evaluation should measure repeated count windows for both known infrared classes with the rock in its realistic covered configuration. A suitable test is to record at least 30 windows for each class at the intended detection distance, calculate the measured count mean and standard deviation, and compare the distributions against the chosen threshold. The subsystem passes if the two measured distributions remain separated with margin under normal rover approach angles. This directly evaluates the actual requirement—classification of $312\,\mathrm{s}^{-1}$ versus $547\,\mathrm{s}^{-1}$—rather than only proving that isolated pulses can be seen on an oscilloscope.

1.3.6 Limitations and further improvements

A further improvement would be an adaptive threshold. A fixed comparator threshold is simple and fast, but the lab lighting level and sensor alignment can change during driving. A slowly varying baseline estimate, either in analogue hardware or software, would allow the threshold to follow ambient conditions while still responding to the short pulse events. This would improve robustness without changing the downstream classification algorithm.

1.4 Magnetic Polarity Detection Subsystem

The magnetic subsystem determines the binary magnetic orientation of each artificial rock, reported by the rover as *Up* or *Down*. This is a top-level classification requirement because the project brief defines magnetic polarity as one of the three rock-type discriminators, alongside infrared pulse rate and ultrasound response. The subsystem was therefore designed as an interacting hardware–software component rather than a standalone demonstration sensor: it must detect polarity at rover operating distance, reject ambient and self-generated interference, and provide a reliable digital label to the central classification logic.

1.4.1 Requirements Derived from the Brief

Table 4 translates the project-level magnetic requirement into measurable subsystem requirements.

Table 4: Magnetic subsystem requirements and current evidence status.

ID	Requirement	Verification metric	Current status
M1	Determine whether the rock magnet is *Up* or *Down*.	Correct polarity label on a known calibration magnet and on all demonstration rocks.	Bench polarity detection shown.
M2	Operate without requiring physical contact with the rock casing.	Valid polarity reading at practical rover approach distance.	Valid readings observed at 5 cm vertical separation and 1.5 cm horizontal offset; final rover-mounted envelope **TBD**.
M3	Reject ambient magnetic field and invalid sensor values.	Baseline-subtracted reading exceeds a fixed noise threshold; out-of-range values are discarded.	Ambient B_z measured at approximately 22 µT; overflow-like values observed and must be rejected.
M4	Integrate electrically with the rover controller.	Stable I^2C communication at 3.3 V; no bus contention with other peripherals.	Bench integration with Adafruit Metro M0 Express complete..

1.4.2 Field-Strength Estimate and Sensing Envelope

A first-order simulation was used to estimate whether a small embedded magnet could be detected at rover-relevant distances. The model assumed an N35–N45 cylindrical magnet of diameter 8 mm and thickness 3 mm. The simulated field falls rapidly with distance, as shown in Table 5. This justified a sensor-placement requirement: the magnetometer should be placed as close to vertically above the rock as the rover mechanics allow, because the useful signal margin decreases sharply beyond approximately 75 mm.

Table 5: Simulated magnet field strength versus sensor distance. Values in gauss are retained for comparison with magnetic-sensor data sheets; 1 G = 100 μT.

Distance	Field /G	Field /μT	Design implication
30 mm	9.11	911	Strong signal; comfortably above ambient.
50 mm	2.11	211	Suitable target working distance.
75 mm	0.646	64.6	Comparable to the Earth's total field; low margin unless baseline-subtracted.
100 mm	0.277	27.7	Poor margin; not a robust design target.

The Earth's magnetic field is on the same order as the weakest simulated target signals. The

design therefore does not rely on absolute field magnitude. Instead, it measures a local baseline before rock approach and classifies the sign of the baseline-subtracted vertical component, $\Delta B_z = B_z - B_{z,0}$. This is more robust than comparing raw readings against a fixed global threshold, because the rover orientation and local ferromagnetic environment can change the measured background.

1.4.3 Sensor Selection

The selected sensor was the DFRobot SEN0419 breakout, based on the Bosch BMM150 three-axis magnetometer. Its relevant specifications are a 0.3 µT resolution, I^2C/SPI digital interface, 3.3 V operation, and magnetic range of ±1300 µT on the x, y axes and ±2047 µT on the z axis. Table 6 shows why this was selected over simpler magnetic switches.

Table 6: Magnetic sensor trade-off.

Candidate	Sensitivity or resolution	Polarity information	Approx. cost	Decision
Hall-effect switch	Approximately 35 G threshold	Binary, but only after a strong field threshold is crossed	£2.55 for 5	Rejected: threshold is too high for the simulated 3 cm–5 cm operating range.
Magnetoresistive sensor	Approximately 25 G threshold	Insufficient for reliable Up/Down classification in the proposed configuration	£10.00	Rejected: high cost and does not provide the required polarity evidence.
SEN0419/BMM150 three-axis magnetometer	0.3 µT resolution	Signed x, y, z vector components	£5.06	Selected: high signal margin at the target range and direct measurement of vertical polarity.

This selection also improves report-level evaluation quality: it produces continuous quantitative data rather than only a binary switch output, allowing the design to be calibrated, plotted, thresholded, and debugged against measurable noise sources.

1.4.4 Electrical and Software Implementation

The SEN0419 was integrated with the Adafruit Metro M0 Express using I^2C. The four active connections used in the bench prototype are shown in Table 7. The breakout is a 3.3 V device, so the sensor is powered from the controller's 3.3 V rail and not from 5 V. The DFRobot `DFRobot_BMM150` library was used to read the geomagnetic vector via `getGeomagneticData()`, returning signed x, y, and z field components in μT.

Table 7: Bench wiring between SEN0419 and Adafruit Metro M0 Express.

SEN0419 pin	Adafruit Metro M0 Express connection
VCC	3.3 V
GND	GND
SDI	SDA
SCK	SCL

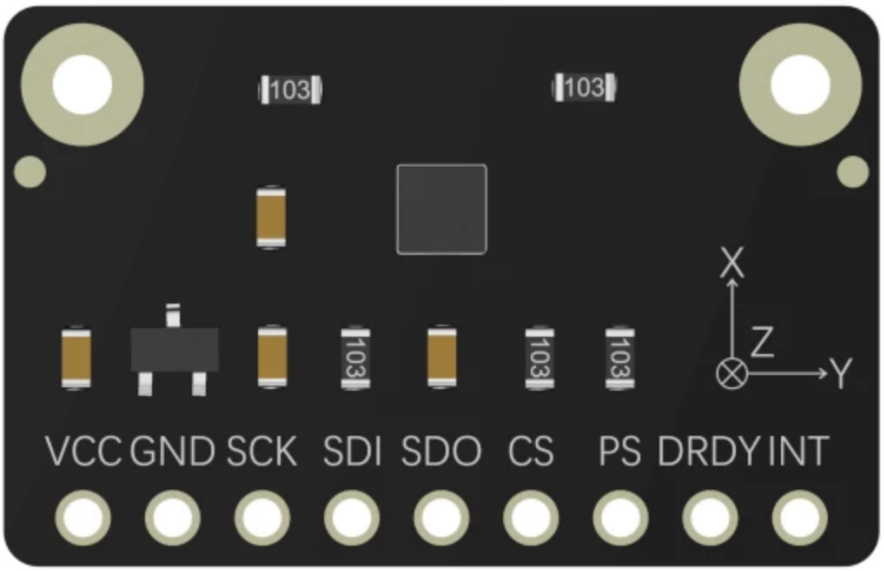


Figure 22: SEN0419/BMM150 breakout showing axis convention and I/O pins.

I^2C was suitable because the sensor data are low-bandwidth compared with the controller clock and the bus provides deterministic master-controlled transactions. For classification, the rover only requires a robust polarity decision rather than high-rate waveform capture.

1.4.5 Bench Testing and Current Evidence

The current bench tests verify that the sensor can measure a meaningful change in vertical field and that polarity can be distinguished under controlled conditions. A magnet was placed approximately 4 cm below the magnetometer, as shown in Figure 23. The observed vertical reading was approximately 130 µT with the magnet present, while the ambient vertical reading was approximately 22 µT. Baseline subtraction therefore gives a target contribution of roughly 108 µT, which is consistent with the simulated order of magnitude at this distance.

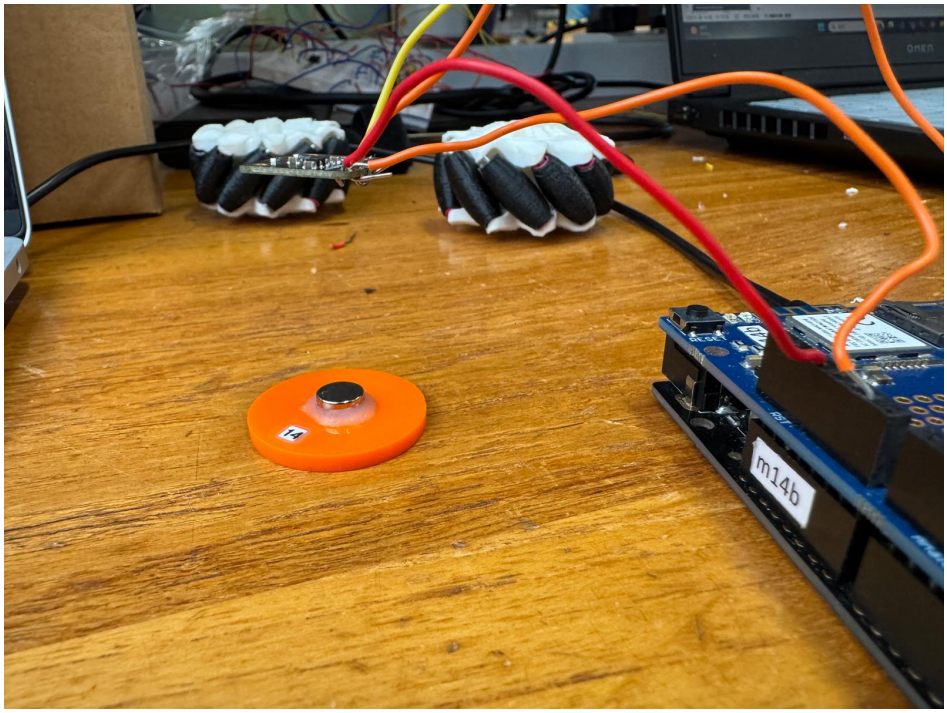


Figure 23: Bench test setup with magnet positioned below the SEN0419 magnetometer.

```
X: 4294967272 uT | Y: 4294967294 uT | Z: 130 uT  → Field UP
X: 4294967272 uT | Y: 4294967294 uT | Z: 128 uT  → Field UP
X: 4294967271 uT | Y: 4294967294 uT | Z: 130 uT  → Field UP
X: 4294967272 uT | Y: 4294967294 uT | Z: 130 uT  → Field UP
X: 4294967272 uT | Y: 4294967294 uT | Z: 129 uT  → Field UP
X: 4294967272 uT | Y: 4294967294 uT | Z: 129 uT  → Field UP
X: 4294967272 uT | Y: 4294967294 uT | Z: 129 uT  → Field UP
X: 4294967271 uT | Y: 4294967294 uT | Z: 129 uT  → Field UP
X: 4294967271 uT | Y: 4294967295 uT | Z: 129 uT  → Field UP
```

(a) Representative field-present readings: $B_z \approx 128\,\mu\text{T}$–$130\,\mu\text{T}$.

```
X: 4294967293 uT | Y: 2 uT | Z: 22 uT  → Weak/ambient
X: 4294967293 uT | Y: 2 uT | Z: 22 uT  → Weak/ambient
X: 4294967293 uT | Y: 2 uT | Z: 22 uT  → Weak/ambient
X: 4294967293 uT | Y: 1 uT | Z: 22 uT  → Weak/ambient
X: 4294967294 uT | Y: 2 uT | Z: 23 uT  → Weak/ambient
X: 4294967294 uT | Y: 2 uT | Z: 23 uT  → Weak/ambient
X: 4294967294 uT | Y: 2 uT | Z: 23 uT  → Weak/ambient
X: 4294967294 uT | Y: 2 uT | Z: 23 uT  → Weak/ambient
X: 4294967294 uT | Y: 2 uT | Z: 23 uT  → Weak/ambient
X: 4294967293 uT | Y: 2 uT | Z: 23 uT  → Weak/ambient
```

(b) Representative ambient readings: $B_z \approx 22\,\mu\text{T}$–$23\,\mu\text{T}$.

Figure 24: Quantitative evidence for baseline subtraction.

A separate polarity test also showed that the sign convention can change consistently when the magnet orientation is reversed. However, Figure 25 shows overflow-like values in one test condition. These values are far outside the BMM150 physical range and must be treated as invalid software or communication artefacts, not as magnetic measurements. This is an important implementation finding: without range checking, a single overflow sample could dominate a threshold rule and cause a false classification.

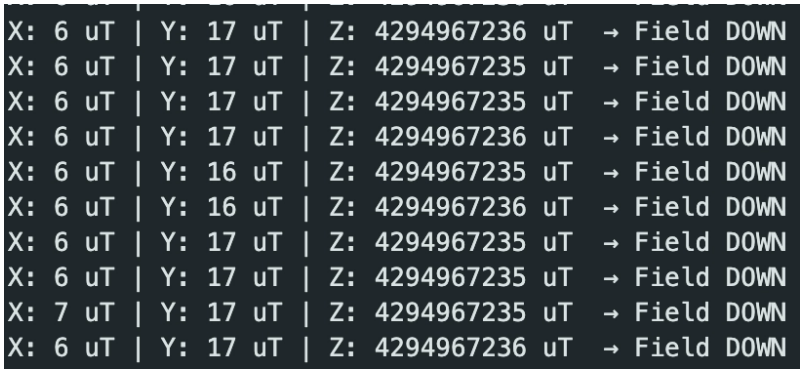


Figure 25: Overflow-like values observed during a polarity test. Final firmware must reject values outside the sensor range before applying the polarity decision.

Table 8 defines the final evidence needed for a high-quality evaluation. The emphasis is on testing both subsystem behaviour and top-level rover behaviour, matching the report marking criterion for evaluation of individual subsystems and the complete system.

Table 8: Magnetic subsystem test matrix

Test	Method	Metric	Outcome
Ambient baseline	Record B_x, B_y, B_z for at least 10 s with no target rock nearby.	Mean, standard deviation, and peak-to-peak noise for each axis.	Initial $B_z \approx 22\,\mu\text{T}$.
Distance response	Measure ΔB_z at fixed vertical separations from 20 mm to 100 mm.	Detection margin $\lvert\Delta B_z\rvert/\sigma_{B_z}$; maximum reliable distance.	5 cm vertical detection observed.
Lateral offset	Sweep sensor laterally across the magnet at the selected probe height.	Maximum offset giving stable Up/Down classification.	1.5 cm horizontal detection observed.
Polarity confusion matrix	Test known Up and Down magnets over repeated trials.	Accuracy, false-Up rate, false-Down rate, no-read rate.	No false.
Rover-mounted test	Repeat distance and polarity tests with the sensor mounted on the final chassis or probe.	Change in detection envelope relative to bench test.	**TBD**.
Motor interference	Repeat ambient and polarity tests with motors idle, turning, and stalled/near-stalled.	Change in σ_{B_z}, threshold margin, and classification accuracy.	**TBD**.
Radio-receiver interference	Compare readings with the 89 kHz resonant receiver disabled and enabled.	Difference in mean B_z, noise, and classification accuracy.	Pull rate 10Hz, low pass filter implanted, no interference

Test	Method	Metric	Outcome
Top-level rock classification	Classify all rocks using the full infrared, ultrasonic, radioage, and magnetic pipeline.	Number of rocks correctly classified; time per rock.	**TBD**.